

Supervised Learning Assignment Proposal: Research Questions, Expected Results, and Result Templates

Student Proposal

Project Title: Predicting NBA Game Outcomes from Player Performance Statistics Using Supervised Machine Learning

Student Name: Aryaman Negi

Student ID: 28147543

Course Name: Machine Learning 120B

Instructor: Prof. Raha Hashim Ali

Dataset Source Link: <https://www.kaggle.com/datasets/eduardopalmieri/nba-player-stats-season-2425>

Purpose. This proposal adapts the supervised learning assignment template to an NBA player statistics dataset. The project uses a recent tabular sports dataset to predict NBA game outcomes from player-level box score performance statistics.

Replaced placeholders. [Dataset Name] = NBA Daily Leaders / NBA Player Stats 2024-25; [Target Variable] = Result; [Application Domain] = NBA player performance and sports analytics; [Models] = Logistic Regression, Decision Tree, k-NN, Random Forest, and Gradient Boosting.

Metric note. Because the project is a binary classification task, the evaluation metrics are Accuracy, Precision, Recall, F1-score, AUC, and confusion matrix.

Dataset and Problem Summary

The dataset file used in this project is nba_dailyleaders_full_24_25.csv. Each row represents one NBA player performance in one game. The dataset contains 28,265 rows and 27 columns. The target variable is Result, which is encoded as W = 1 for win and L = 0 for loss. The project is therefore a binary classification problem.

Dataset Summary

Item	Value
Dataset name	NBA Daily Leaders / NBA Player Stats 2024-25
Dataset file	nba_dailyleaders_full_24_25.csv
Rows	28,265
Columns	27
Target variable	Result
Task type	Binary classification
Positive class	W = Win
Negative class	L = Loss

Class Distribution

Class	Count
W	14206
L	14059

RQ1. Baseline Performance

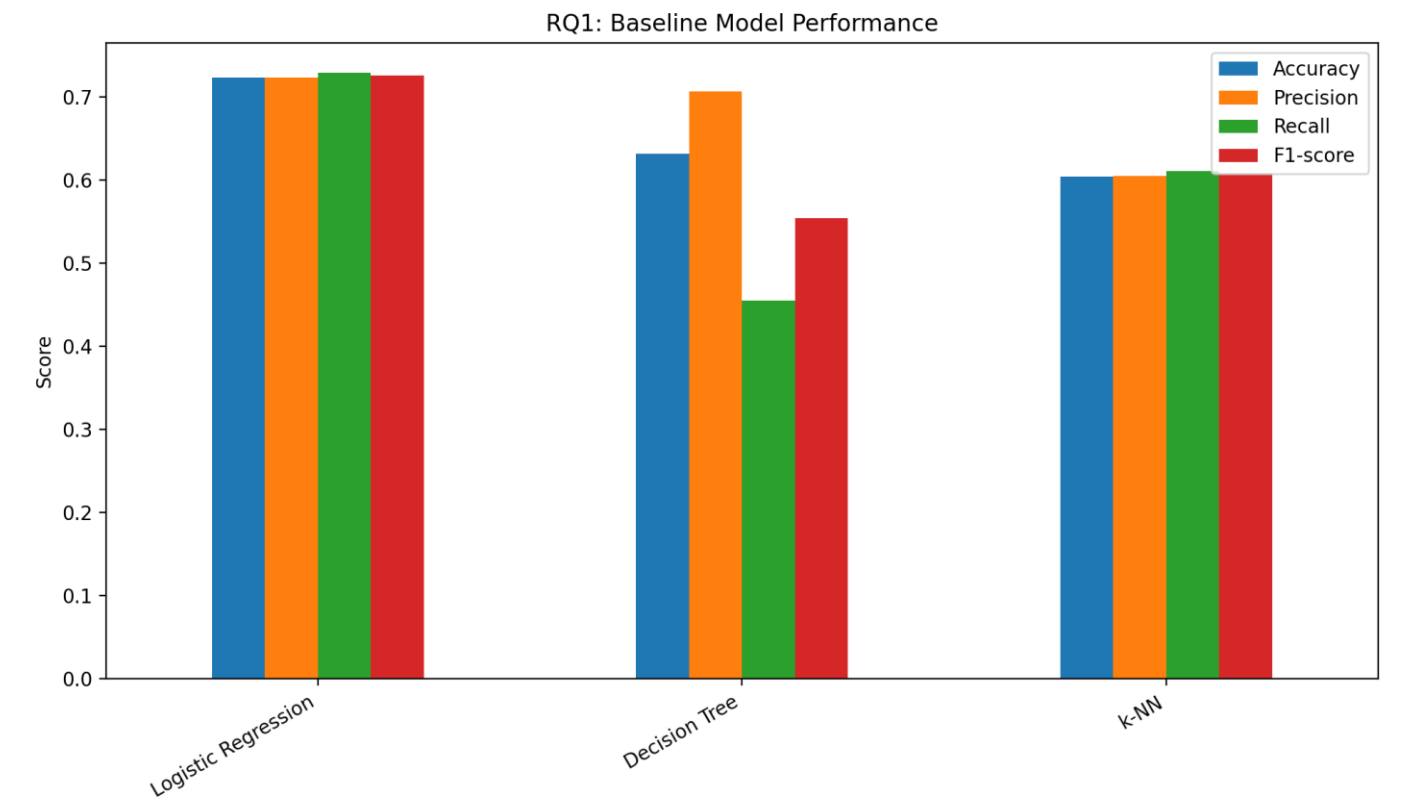
Research Question. How effectively can baseline supervised learning models predict NBA game outcomes using individual player performance statistics from the NBA Daily Leaders / NBA Player Stats 2024-25 dataset?

Expected Results. Baseline models are expected to achieve reasonable but not optimal performance, confirming that the dataset contains useful predictive patterns for distinguishing wins from losses.

Table I. Baseline model performance on NBA Daily Leaders / NBA Player Stats 2024-25

Model	Accuracy	Precision	Recall	F1-score	AUC	Train Time (s)
Logistic Regression	0.7237	0.7234	0.7290	0.7262	0.7898	0.4032
Decision Tree	0.6315	0.7071	0.4555	0.5541	0.6731	0.2410
k-NN	0.6045	0.6054	0.6114	0.6084	0.6406	0.0957

Figure 1. Baseline performance comparison of candidate supervised learning models on the NBA dataset. Suggested figure type: grouped bar chart.



RQ2. Model Comparison

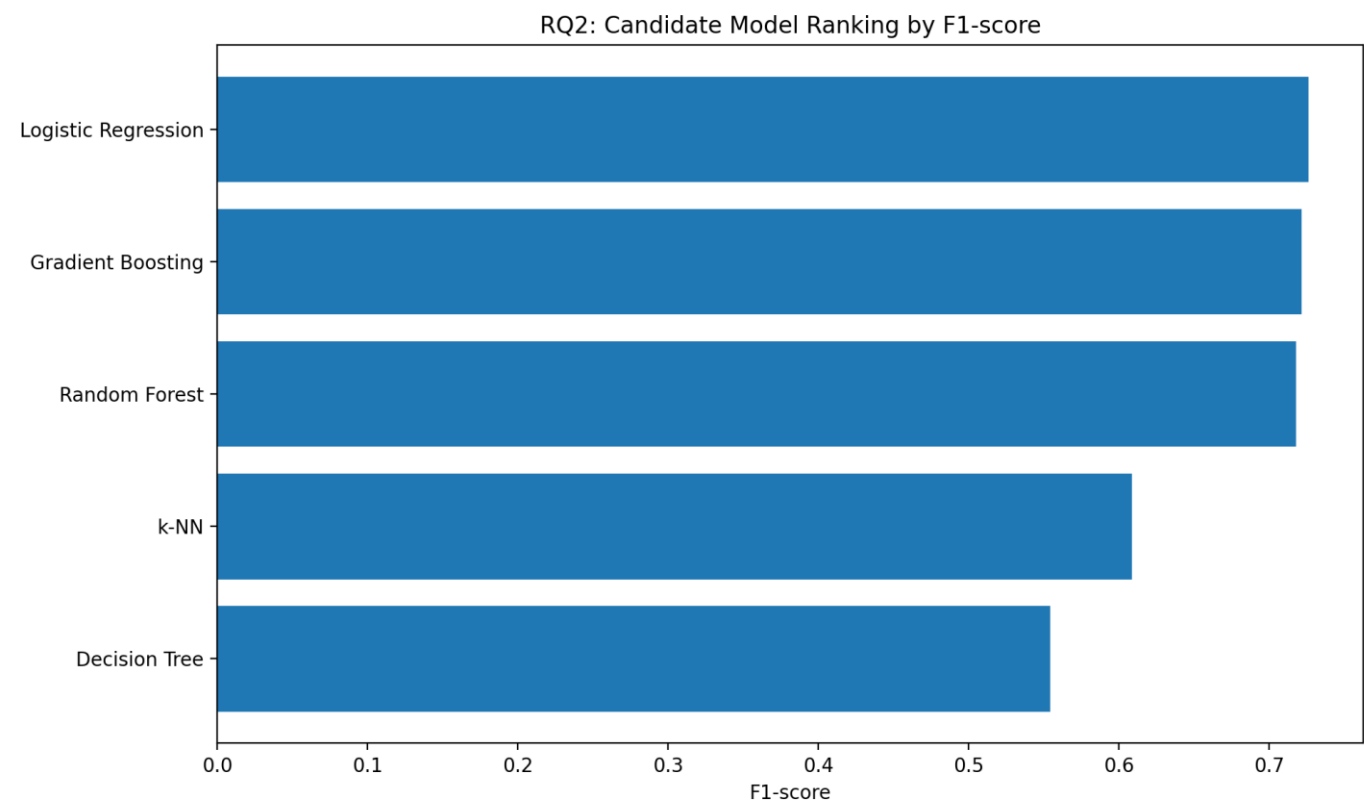
Research Question. Which supervised learning model achieves the best predictive performance for NBA game outcome prediction, and how does it compare with competing methods?

Expected Results. Advanced or ensemble-based models are expected to compete strongly with simpler baselines. However, the best model may depend on the selected evaluation metric because Accuracy, F1-score, and AUC can rank models differently.

Table II. Comparative performance of candidate supervised learning models

Model	Accuracy	Precision	Recall	F1-score	AUC	Train Time (s)
Logistic Regression	0.7237	0.7234	0.7290	0.7262	0.7898	0.4032
Gradient Boosting	0.7223	0.7269	0.7166	0.7217	0.7936	3.6022
Random Forest	0.7148	0.7139	0.7219	0.7179	0.7973	3.3030
k-NN	0.6045	0.6054	0.6114	0.6084	0.6406	0.0957
Decision Tree	0.6315	0.7071	0.4555	0.5541	0.6731	0.2410

Figure 2. Overall ranking of candidate models on the NBA dataset using primary evaluation metrics. Suggested figure type: horizontal bar chart or radar plot.



RQ3. Effect of Preprocessing

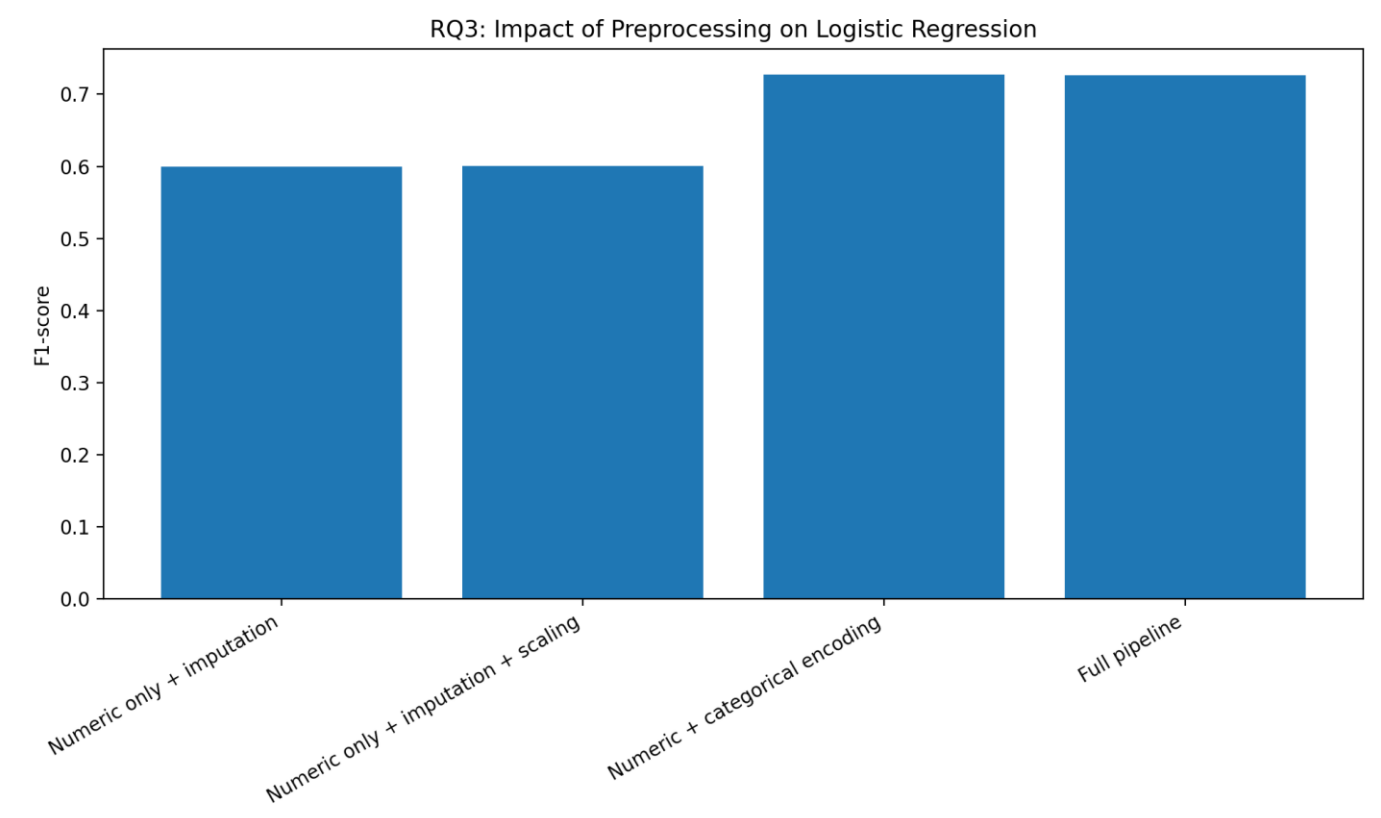
Research Question. How do different data preprocessing strategies affect the performance of supervised learning models on the NBA player statistics dataset?

Expected Results. Preprocessing steps such as missing-value imputation, minutes conversion, numerical scaling, and categorical encoding are expected to improve predictive performance, with categorical team and opponent information providing additional predictive value.

Table III. Impact of preprocessing strategies on model performance

Preprocessing Strategy	Accuracy	Precision	Recall	F1-score	AUC
Numeric only + imputation	0.5857	0.5831	0.6163	0.5992	0.6233
Numeric only + imputation + scaling	0.5864	0.5837	0.6174	0.6001	0.6232
Numeric + categorical encoding	0.7244	0.7241	0.7297	0.7269	0.7898
Full pipeline	0.7237	0.7234	0.7290	0.7262	0.7898

Figure 3. Performance gains achieved by different preprocessing strategies. Suggested figure type: ablation bar chart or line plot.



RQ4. Feature Importance and Interpretability

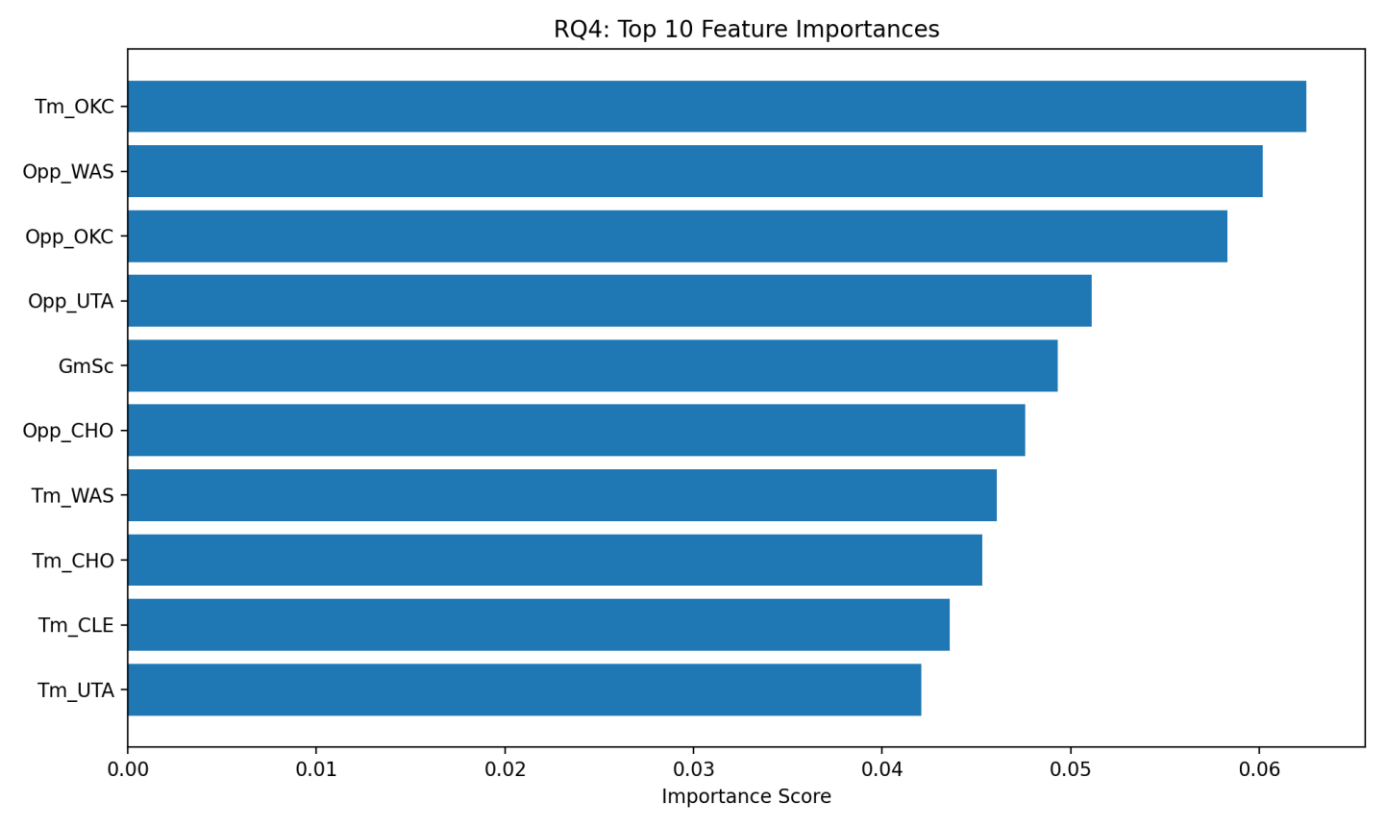
Research Question. Which input features contribute most to predictive performance, and what basketball insights can be derived from the most influential variables?

Expected Results. A subset of player statistics and contextual features is expected to contribute disproportionately to predictive performance. The main model excludes +/- to reduce outcome leakage because plus/minus is closely related to the final game result.

Table IV. Top-ranked features contributing to model prediction

Rank	Feature	Importance Score	Direction / Effect	Domain Interpretation
1	Tm_OKC	0.0625	Team context signal	Team identity captures team strength and style context.
2	Opp_WAS	0.0602	Opponent context signal	Opponent identity captures matchup difficulty.
3	Opp_OKC	0.0583	Opponent context signal	Opponent identity captures matchup difficulty.
4	Opp_UTA	0.0511	Opponent context signal	Opponent identity captures matchup difficulty.
5	GmSc	0.0493	Positive performance signal	Overall player game score helps separate wins and losses.

Figure 4. Feature importance analysis of the best-performing model. Suggested figure type: top-10 feature importance bar plot or SHAP summary plot.



RQ5. Sensitivity to Evaluation Metrics

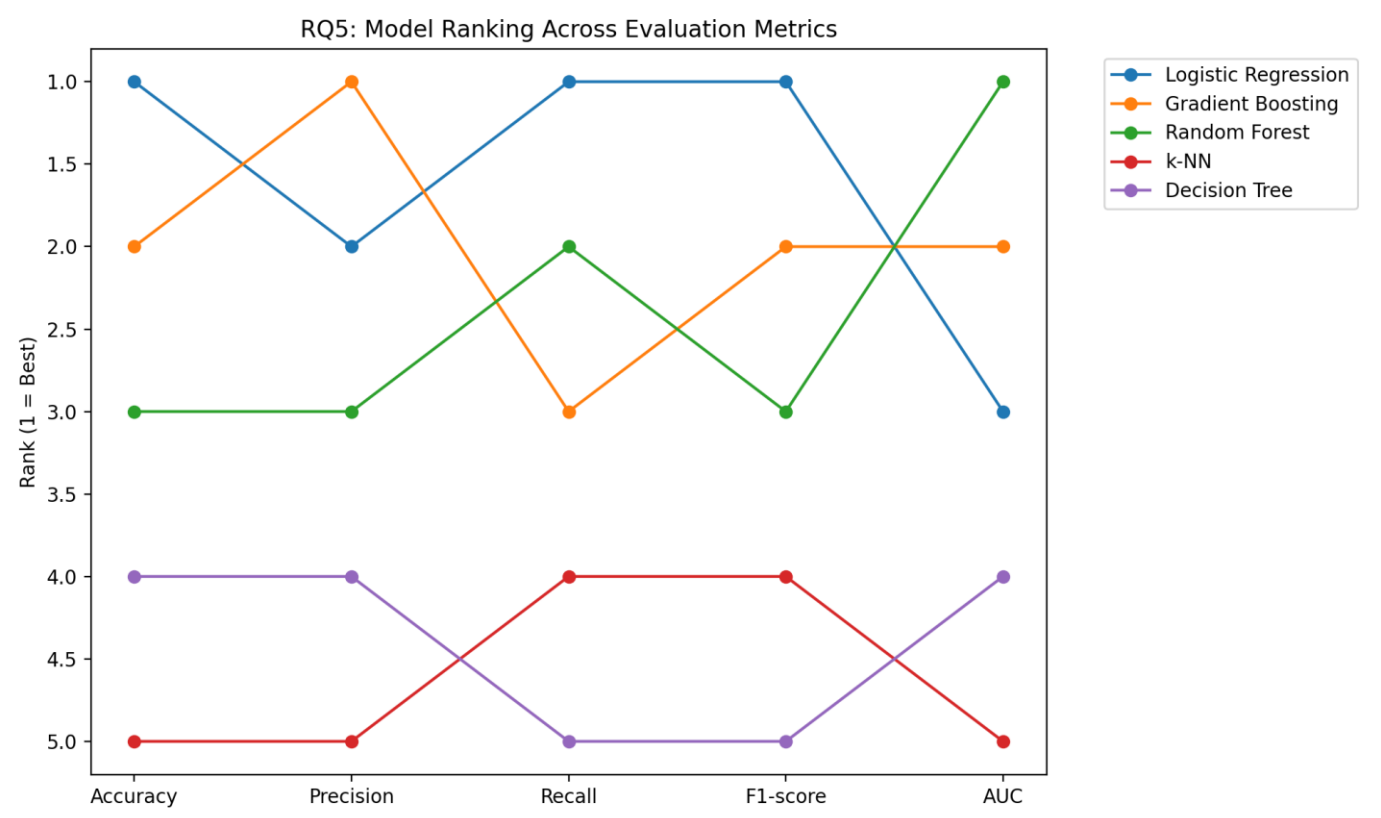
Research Question. How does the relative ranking of candidate models change when different evaluation metrics are considered?

Expected Results. Model rankings are expected to vary across metrics, showing that metric choice can influence which model appears best. In this project, F1-score, AUC, and Accuracy are all reviewed to support a balanced final recommendation.

Table V. Model ranking under different evaluation metrics

Model	Rank by Accuracy	Rank by Precision	Rank by Recall	Rank by F1-score	Rank by AUC
Logistic Regression	1	2	1	1	3
Gradient Boosting	2	1	3	2	2
Random Forest	3	3	2	3	1
k-NN	5	5	4	4	5
Decision Tree	4	4	5	5	4

Figure 5. Variation in model rankings across different evaluation metrics. Suggested figure type: slope graph or bump chart.



RQ6. Robustness and Generalization

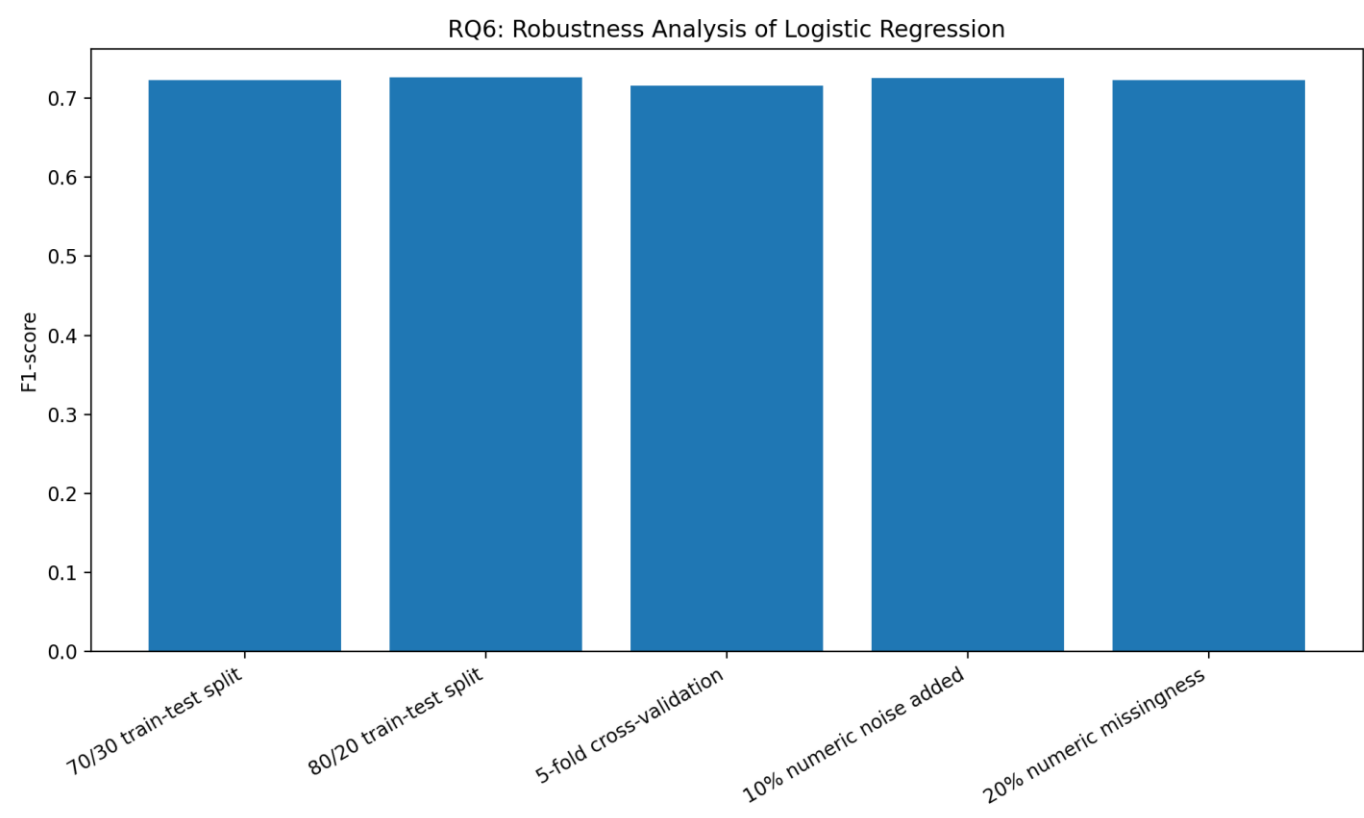
Research Question. How robust is the selected supervised learning model under different train-test splits, cross-validation settings, or data perturbation scenarios?

Expected Results. The selected model is expected to maintain relatively stable performance across validation settings and degrade gradually under noisy or incomplete data conditions.

Table VI. Robustness analysis of the selected model under varying experimental conditions

Scenario	Accuracy	Precision	Recall	F1-score	AUC	Std. Dev.
70/30 train-test split	0.7208	0.7212	0.7245	0.7228	0.7840	
80/20 train-test split	0.7237	0.7234	0.7290	0.7262	0.7898	
5-fold cross-validation	0.7142			0.7155		0.0070
10% numeric noise added	0.7237	0.7242	0.7272	0.7257	0.7893	
20% numeric missingness	0.7198	0.7194	0.7254	0.7224	0.7827	

Figure 6. Robustness of the selected model under cross-validation and perturbed data conditions. Suggested figure type: boxplot or line chart.



RQ7. Practical Usefulness and Final Recommendation

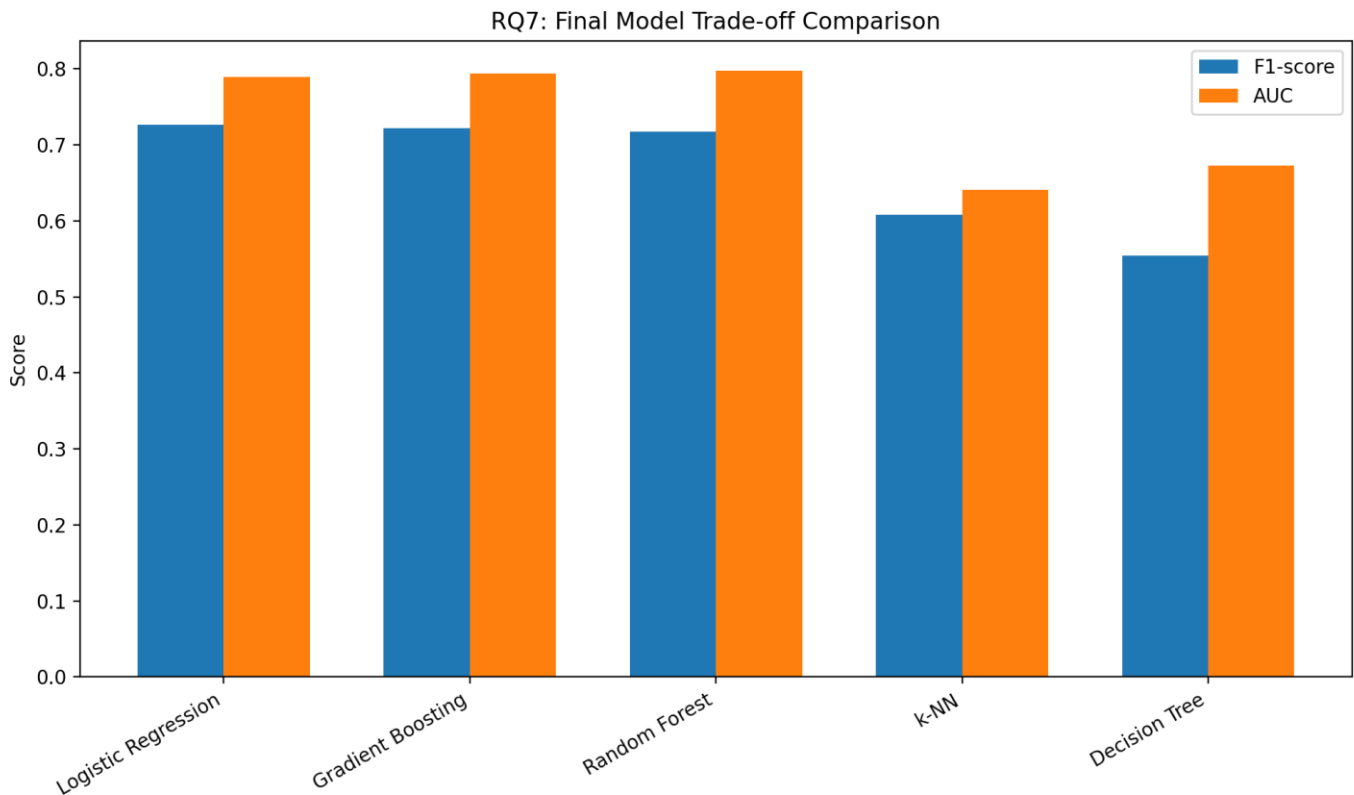
Research Question. To what extent is the developed supervised learning solution practically useful, interpretable, and reliable for decision-making in NBA sports analytics?

Expected Results. The final model is expected to provide a balanced trade-off among predictive performance, interpretability, robustness, computational cost, and deployment practicality.

Table VII. Final comparative decision matrix for model selection

Model	Predictive Performance	Interpretability	F1-score	AUC	Computational Cost	Final Recommendation
Logistic Regression	High	High	0.7262	0.7898	Low	Selected
Gradient Boosting	High	Medium-Low	0.7217	0.7936	High	Not selected
Random Forest	High	Medium	0.7179	0.7973	High	Not selected
k-NN	Medium	Low	0.6084	0.6406	Low	Not selected
Decision Tree	Low	High	0.5541	0.6731	Low	Not selected

Figure 7. Final trade-off analysis among predictive performance, interpretability, and robustness. Suggested figure type: radar chart or bubble chart.



Suggested Expected Results Section

- Baseline models demonstrate that the NBA player statistics dataset is suitable for supervised learning.
- Ensemble models such as Random Forest and Gradient Boosting are expected to be competitive on tabular data, while Logistic Regression may remain strong due to the structured nature of the features.
- Proper preprocessing improves predictive performance, especially when categorical team, opponent, and home/away variables are encoded.
- Feature importance analysis identifies the most influential variables and provides basketball-related interpretation.
- Model rankings vary under different evaluation metrics, so the final recommendation considers multiple metrics instead of one score only.
- The best-performing model remains reasonably stable under different validation settings and data perturbation tests.
- The final selected model provides a practical balance between performance, interpretability, robustness, and reliability.

How This Proposal Adapts the Handout

- The placeholder [Dataset Name] is replaced with NBA Daily Leaders / NBA Player Stats 2024-25.
- The placeholder [Target Variable] is replaced with Result.
- The placeholder [Application Domain] is replaced with NBA player performance and sports analytics.
- Because the task is classification, the proposal reports Accuracy, Precision, Recall, F1-score, AUC, and confusion matrix.
- The dummy tables and figure ideas are adapted to the actual NBA dataset and match the executed notebook outputs.

Conclusion

This proposal presents a supervised learning project for predicting NBA game outcomes using player-level game statistics. The target variable is Result, and the task is binary classification. The project compares multiple supervised learning models, evaluates preprocessing strategies, analyzes feature importance, tests model robustness, and recommends a final model for practical NBA sports analytics use.